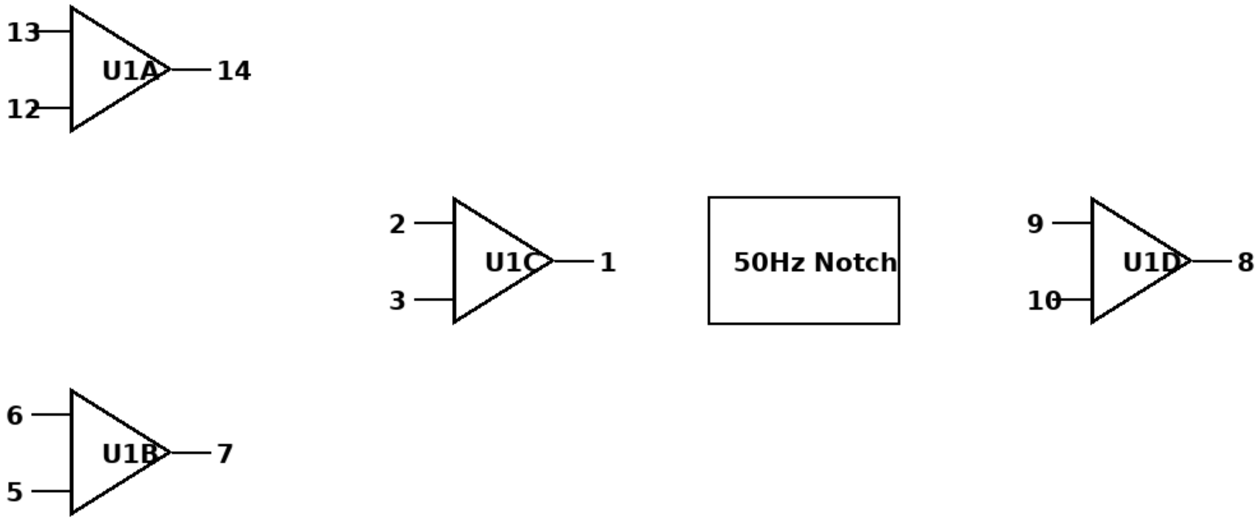
DIY EEG Circuit Guide (TL074 + Single 9V Battery)

EEG Circuit Schematic (Single TL074)



VCC: 9V Battery
REF: 4.5V VGND

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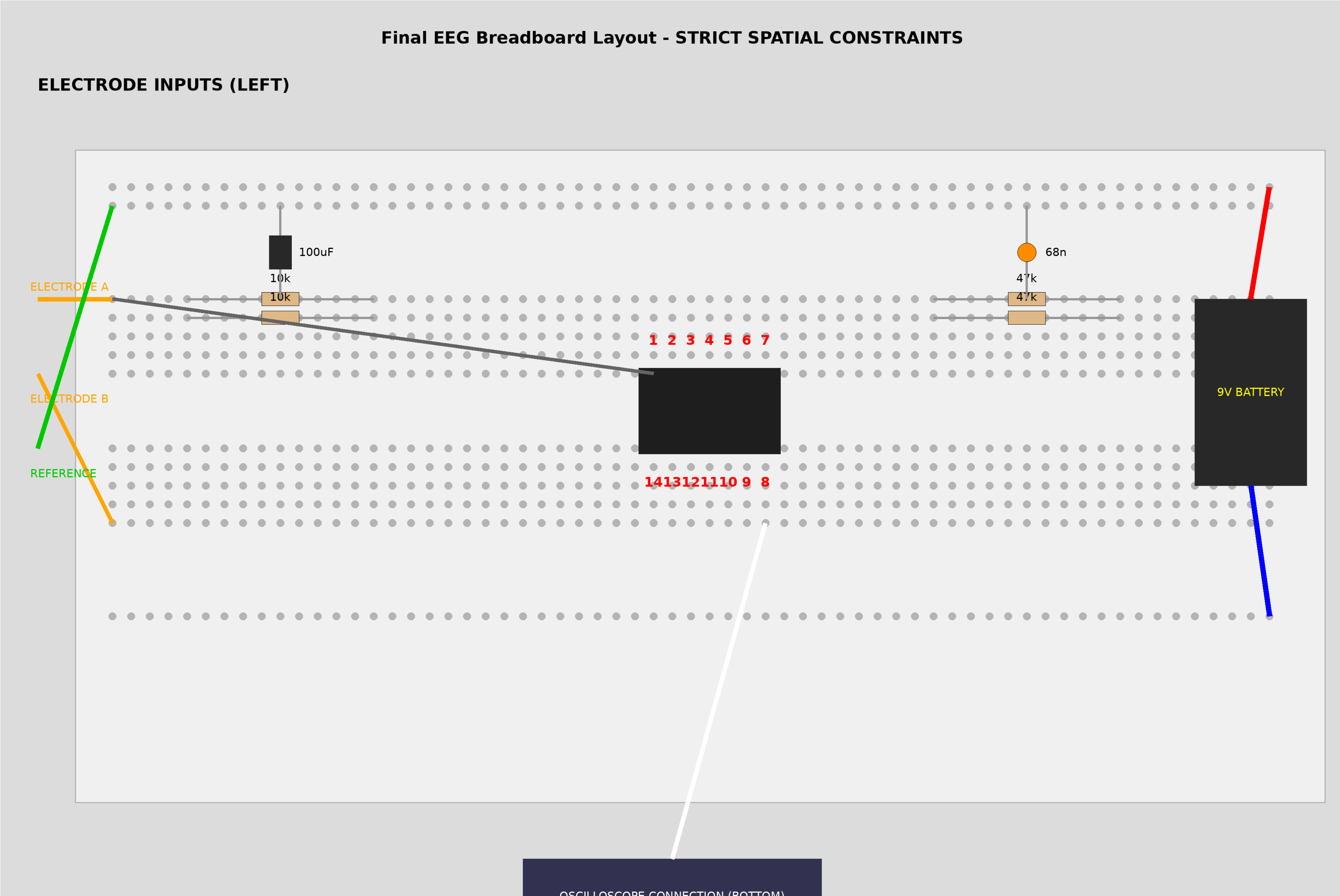

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DESIGN

EEG Circuit Design (Single TL074, Single 9V Battery)

Architecture

1. Virtual Ground (VGND): 4.5V reference using two 10k resistors and a 100uF capacitor.

2. Stage 1: Instrumentation Amplifier: 3-op-amp InAmp, Gain ~21.

3. Stage 2: 50Hz Notch Filter: Passive Twin-T, tuned for 49.8Hz.

4. Stage 3: Gain + Bandpass Filter: Gain ~471, Bandpass 3.4Hz - 34Hz.

Total Gain: ~10,000.

Component List

- 1x TL074CN IC

- 1x 9V Battery

- Resistors: 2x 1k, 8x 10k, 4x 47k, 1x 470k, 1x 1M.

- Capacitors: 4x 68n, 1x 10n, 1x 100n, 1x 47uF, 1x 100uF.

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ASSEMBLY

Breadboard Assembly Guide

Step 1: Power & VGND

- Red Rail: +9V. Blue Rail: 0V.
- VGND: Junction of two 10k resistors between +9V and 0V. Stabilize with 100uF to 0V.

Step 2: TL074 IC

- Side 1 (Bottom): Pins 1-7. Side 2 (Top): Pins 14-8.
- Pin 4 to +9V. Pin 11 to 0V.

Step 3: Stages

- Stage 1 (InAmp): Pins 12, 13, 14, 5, 6, 7, 2, 3, 1.
- Stage 2 (Notch): Pin 1 to Input, Output to Pin 10.
- Stage 3: Output at Pin 8.

Step 4: Connections

- Electrodes: Left side of board.
- Battery: Right side of board.
- Oscilloscope: Bottom of board.

TESTING

Safety & Testing

Safety

1. Battery Only.
2. Oscilloscope Isolation.
3. Reference Grounding.

Testing

- Check VGND (4.5V).
- Short inputs to VGND, check output (Flat line).
- Capture Alpha waves (8-13Hz) with eyes closed.